

LESSON 6-1 • SESSION 2 • PRACTICE • FORM B

Dividing a Fraction by a Whole Number

Learning target – I can divide a fraction by a whole number and explain the result. · 6.NS.A.1

REMEMBER

- Dividing by a whole number **shares** the fraction into equal parts.
- Multiply by the reciprocal** of the whole number:
 $2/3 \div 4 = 2/3 \times 1/4 = 2/12 = 1/6$.
- Each share is **less** than the fraction. Check: share $\times$ number of shares = the fraction.

WORKED EXAMPLE

A painter pours $2/3$ liter of paint evenly into 4 trays. How much paint is in each tray?

Step 1 Write the division: $2/3 \div 4$.

Step 2 Multiply by the reciprocal: $2/3 \times 1/4 = 2/12$.

Step 3 Simplify: $2/12 = 1/6$. Check: $1/6 \times 4 = 4/6 = 2/3$.

Answer $1/6$ liter in each tray

WORDS reciprocal (*recíproco*) · multiplicative inverse (*inverso multiplicativo*) · divisor (*divisor*)

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. $1/4 \div 2 =$ _____

b. $2/3 \div 4 =$ _____

c. $3/4 \div 3 =$ _____

d. $8/9 \div 4 =$ _____

e. $5/7 \div 5 =$ _____

f. $4/9 \div 8 =$ _____

2. **Kitchen Supplies.** A cooking club splits $5/6$ pound of cheese evenly among 5 cooks, as the tape shows.

5/6 lb

?	?	?	?	?
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a. How much cheese does each cook get? _____ lb

b. They pour $3/4$ gallon of broth evenly into 8 jars. Amount per jar? _____ gal

3. **Clay Balls.** A sculptor has $2/5$ kilogram of clay. He divides it equally into 8 balls. What is the mass of each ball?

Hint Multiply $2/5$ by $1/8$, then simplify.

Show your work.

Answer _____ kg

4. **Glue Bottles.** A jar holds $9/10$ liter of glue. It is poured evenly into 3 bottles. How much glue is in each bottle?

Hint Multiply by $1/3$, then simplify.

Show your work.

Answer _____ L

5. **String Pieces.** A $2/3$ -yard string is cut into 4 equal pieces. Malik says each piece is $8/3$ yards long. Justify or correct his answer.

Hint Is one piece longer or shorter than the whole string?

Show your work.

Answer _____ yard

6. **Juice.** Ana has $\frac{9}{10}$ liter of juice. She pours out $\frac{3}{10}$ liter, then shares the rest equally among 3 friends.

Hint Subtract first, then share the rest.

- a. How much is left? _____ L
b. How much does each friend get? _____ L

Show your work.

7. **Lab Day.** A chemist has $\frac{7}{8}$ liter of solution to split evenly among beakers.

- a. How much goes in each beaker if there are 5 beakers? _____ L
b. How much if there are 10 beakers? _____ L
c. How did doubling the number of beakers change each share? _____

8. **Fabric Strip.** A strip of fabric is $\frac{6}{7}$ meter long. It is cut into 3 equal pieces. Find the length of one piece. Then check by multiplying.

Hint Check: your answer $\times 3$ should equal $\frac{6}{7}$.

Show your work.

Answer _____ m

FIND THE ERROR

explain the mistake, then fix it

9. **Jordan's Work.** Jordan solved $\frac{4}{9} \div 2$. He kept the divisor right side up and wrote $\frac{4}{9} \times \frac{2}{1} = \frac{8}{9}$. Explain his mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $\frac{3}{5} \div 3$.

A $\frac{9}{5}$ **B** $\frac{1}{5}$ **C** $\frac{5}{9}$ **D** $\frac{3}{5}$

11. Dividing a fraction by a whole number greater than 1 makes the result –

A greater than the fraction **B** equal to the fraction
C smaller than the fraction **D** always a whole number

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

1/20 P	1/6 R	7/40 I	16/5 X	8/3 Q	2/7 N	3/10 O	1/5 T	27/10 Z	7/80 O
#3		#4	#5	#6b	#7a	#7b	#8		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____